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midhun@localhost: ~  
midhun@localhost ~]$ ls  
a      bank      Desktop  mail.c   Music    sample.txt  while.sh  
a      banker.c  Documents midhun   Pictures sh          whilew.sh  
ABC    bankers  Downloads mai       Public   Templates  XYZ  
abc.sh  bm       h        dev      REC      test.sh  
ab.sh   close    mai      midhun   midhun.c Videos  
midhun@localhost ~]$ vi bestfit  
midhun@localhost ~]$ vi bestfit.c  
midhun@localhost ~]$ gcc bestfit.c -o bestfit  
midhun@localhost ~]$ ./bestfit
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midhun@localhost ~]$ ./bestfit  
Enter number of processes: 5  
Enter number of resource types: 3  
Enter Allocation Matrix:  
0 1 0  
2 0 0  
3 0 2  
2 1 1  
0 0 2  
Enter Max Matrix:  
7 5 3  
3 2 2  
9 0 2  
2 2 2  
4 3 3  
Enter Available Resources:  
3 3 2  
Safe sequence is: P1 P3 P4 P0 P2  
midhun@localhost ~]$ █
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#include <stdio.h>

int main() {
    int n, r, i, j, k;

    // Input number of processes and resources
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &r);

    int alloc[n][r], max[n][r], need[n][r], avail[r];
    int finish[n], safe_sequence[n];
    int count = 0, safeFound;

    // Allocation Matrix
    printf("Enter Allocation Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < r; j++) {
            scanf("%d", &alloc[i][j]);
        }
    }

    // Max Matrix
    printf("Enter Max Matrix:\n");
    for(i = 0; i < n; i++) {
        for(j = 0; j < r; j++) {
            scanf("%d", &max[i][j]);
        }
    }

    // Available Resources
    printf("Enter Available Resources:\n");
    for(i = 0; i < r; i++) {
        scanf("%d", &avail[i]);
    }
}

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    int safe = 1;

    for(j = 0; j < r; j++) {
        if(need[i][j] > avail[j]) {
            safe = 0;
            break;
        }
    }

    if(safe) {
        for(k = 0; k < r; k++) {
            avail[k] += alloc[i][k];
        }

        safe_sequence[count] = i;
        finish[i] = 1;
        safeFound = 1;
        count++;
    }
}

if(!safeFound) {
    printf("System is in unsafe state! May occur deadlock.\n");
    return 0;
}

// Print Safe Sequence
printf("Safe sequence is: ");
for(i = 0; i < n; i++) {
    printf("%d ", safe_sequence[i]);
}
printf("\n");

return 0;
}

```